

Curriculum Vitae

Cubaka ZAHINDA René

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Summary

Born November 15th, 1999, in the South-Kivu Region in the Democratic Republic of the Congo. My academic expertise lies in solving real-life problems using mathematical tools. I enjoy research in both applied and pure mathematics and have worked in both during my BSc and master's studies.

I have over five years of experience teaching and tutoring students at both secondary and university levels. My approach focuses on breaking down complex concepts into understandable steps, identifying each student's unique learning style, and fostering a genuine appreciation for the subject.

I hold a master's degree from the African Institute for Mathematical Sciences (AIMS Ghana), and I want to be not only a lecturer, but a lecturer who shows the way to students and leads them to loving mathematics.

My goal is to do a PhD in mathematics because it will help me to solve real-life problems using mathematical tools.

Education

African Institute for Mathematical Sciences	Accra
<i>Master in Mathematical Sciences, Ghana</i>	<i>Sep. 2024–June 2025</i>
Graduated with a distinction	
Institut Supérieur Pédagogique de Kaziba	South-Kivu
<i>Graduate degree in Mathematics for education, DR Congo</i>	<i>2021–2023</i>
Graduated with Great Distinction	
Institut Supérieur Pédagogique de Kaziba	South-Kivu
<i>Under graduate degree in Mathematics and Physics for education, DR Congo</i>	<i>2018–2021</i>
Graduated with Distinction	

Researches

2025: Master essay: *The length of a group element and the Cayley graph of a group*,
Supervisor: Prof. Dr. Peter Webb, University of Minnesota, USA
African Institute for Mathematical Sciences (AIMS) Ghana

2023: Graduate thesis (Bac+5): *Prediction of Hospital Length of Stay in Gynecology based on Machine Learning: Case of some hospitals in South Kivu*,
Supervisor: Prof. Dr. ZIHINDULA MUSHENGGEZI Elie
Institut Supérieur Pédagogique de Kaziba South-Kivu, DR Congo

2021: Undergraduate thesis (Bac+3): *Mathematical modeling of the population growth*,
Supervisor: Zirhumanana BALIKE Dieudonné
Institut Supérieur Pédagogique de Kaziba South-Kivu, DR Congo

Presentations in conferences

22–23 Avril 2026: Session Speaker: Machine Learning Models for Predicting Length of Hospital Stay in the Gynecology Department of South-Kivu Hospitals

Mathematics & Physics Frontiers 2026 – Theories, Models, and Applications, Frankfurt, Germany

2024: Topic: Mathematical modelling of propagation of a rumour on social networks

Institut Supérieur Pédagogique de Kaziba, South-Kivu, DR Congo

2022: Topic: The PageRank method in an online search. The case of the Google search engine

Institut Supérieur Pédagogique de Bukavu South-Kivu DR Congo

Teaching Experience and Positions

January 2024–Present: Research and Teaching Assistant (Junior Lecturer), Institut Supérieur Pédagogique de Kaziba (ISP Kaziba)

South-Kivu, DR Congo

- Courses in Affine geometry taught
- Course in Linear Algebra Taught
- Course in Galois theory taught
- Course Ordinary Differential Equations
- Taught course of Data analysis

2024–Present: Visiting Assistant at Institut Supérieur de Développement Rural de Kaziba (ISDR Kaziba)

South-Kivu, DR Congo

December 2025: Visiting Assistant at Institut Supérieur Pédagogique de Walungu (ISP Walungu)

South-Kivu, DR Congo I taught course in **Measure Theory** and **Differentiale Geometry**

December 2025: Visiting Assistant at Institut Supérieur des Etudes Commerciales et Financière (ISECOF) de Bukavu

South-Kivu, DR Congo I taught course in **Mathematic Analysis in Economical Science**

January 2024–Present:

Chief of the staff of the General Director of ISP Kaziba South-Kivu DR Congo Served as Chief of Staff, managing the Director's office, coordinating the executive agenda, and overseeing strategic initiatives.

2019–2024:

Teacher of mathematics and Physics at secondary schools South-Kivu DR Congo

Summer Schools, Research Programmes and Workshops

08 Jul.– 08 Sep. 2026:

MIORPA 2026 (Mathematical Institute's Online Research Projects with Africa), University of Oxford, United Kingdom.

Selected for a competitive eight-week research mentorship programme for outstanding pre-PhD students based in Sub-Saharan Africa.

Project: *Computer-Assisted Study of Invariants and Structural Properties of Finite Groups*.

Mentor: **Paula Heim** (University of Oxford).

The project investigates relationships between numerical invariants and structural properties of finite groups using computational algebra (GAP) and data-driven methods, including machine learning techniques for predicting properties such as nilpotency and solvability.

19–21 May 2026:

Conference *Helping Teachers Teach Mathematics* (HTTMC 2026), African Institute for Mathematical Sciences (AIMS Ghana), University of Ghana, Accra, Ghana.

Theme: *Bridging the Gap: Strengthening Mathematics Classrooms in Africa through Resilience, Innovation, and Hope.*

Participation in workshops and sessions focusing on mathematics teaching, problem-solving, innovative pedagogical approaches, and the integration of mathematics and computing in African education.

27th–30th October 2025:

1st AIMS School on The Theory of Quantum Learning Algorithms, African Institute for Mathematical Sciences AIMS South Africa,

This was a School on the theory of quantum algorithms where we lent several topics such as quantum learning theory, the Clifford group, Haar integration, and quantum tomography.

27th–30th October 2025:

1st AIMS Workshop on The Theory of Quantum Learning Algorithms, African Institute for Mathematical Sciences AIMS South Africa,

This was a Workshop on the theory of quantum algorithms where different researchers present their work and some open problems

15th–27th July 2024:

Ecole Mathématique Africaine (EMA), Université de Lomé,

Topic: Mathematical approach to problems and industrial problems

Awards and Certificates

2026: University of Oxford

Selected Participant, MIORPA 2026 (Mathematical Institute's Online Research Projects with Africa), a competitive research mentorship programme for pre-PhD students in Sub-Saharan Africa.

2025: AIMS Ghana

F.A.K. Allotey Meritorious Award for achieving distinction in coursework and thesis (above 85%)

2024: ISP Kaziba

Excellence Prize for being the overall best student of the university

2019: ISP Kaziba

Excellence Prize for being the overall best student of the university

Skills

Programming skills: Python, R, GAP

Software Proficiencies Photoshop, Excel, Word, PowerPoint, LaTeX

Languages English (Intermediate), French (Fluent)

I certify on my honour that the information provided above is true and accurate

Cubaka ZAHINDA René